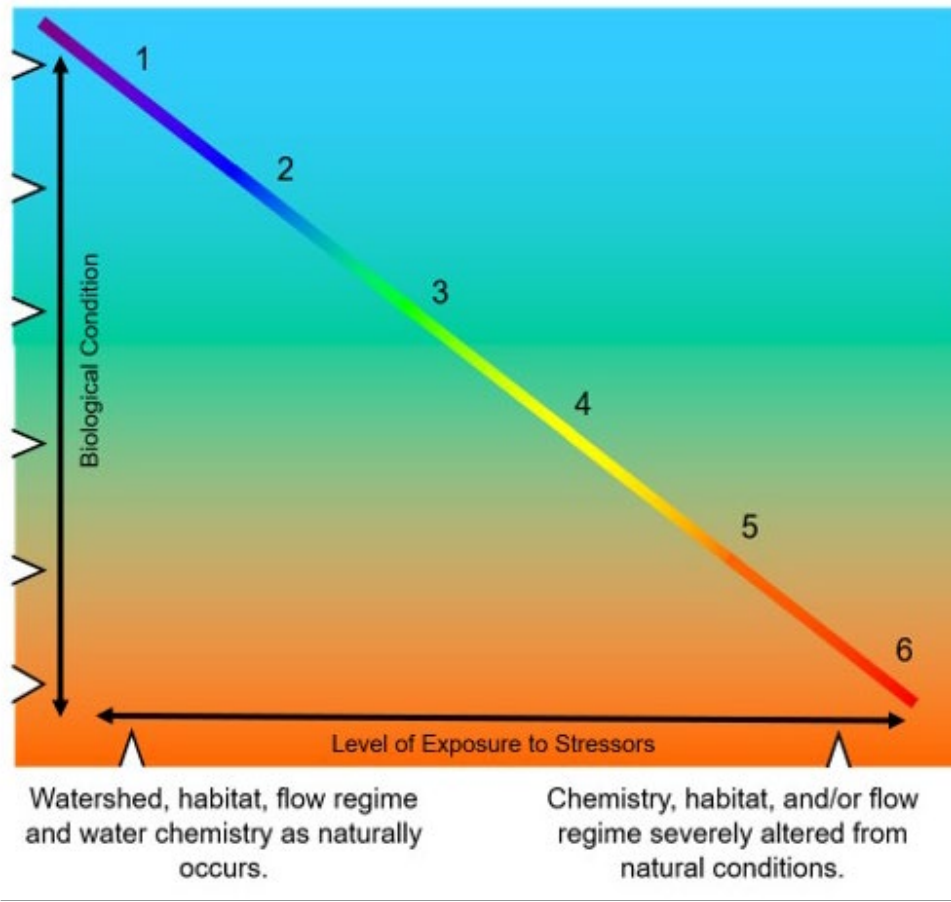




# Quick tutorial: Lower Boise River Shiny app BCG calculator

[tetrattech-wtr-wne.shinyapps.io/LowerBoiseBCGCalc/](https://tetrattech-wtr-wne.shinyapps.io/LowerBoiseBCGCalc/)

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This R-based tool calculates Biological Condition Gradient (BCG) scores for macroinvertebrate and fish samples from the Lower Boise River mainstem.



# Requirements

Users do not need R on their computers to run the Shiny app, nor do they need to have any familiarity with R.

To run the Shiny app you need:

- Internet access\*
- Intermediate-level computer skills (need to be able to browse, upload and download files, unzip files)

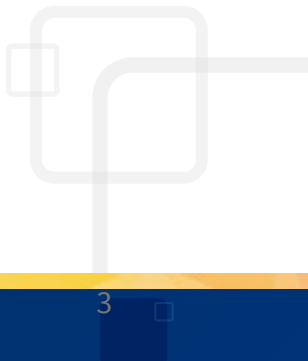
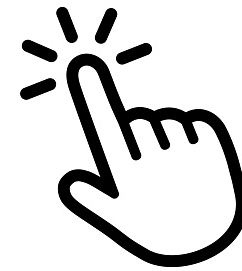
For those who prefer to work with R, the functions are carried out by the [BCGcalc](#) and [BioMonTools](#) R packages, which are available on GitHub and were developed by Erik W. Leppo from Tetra Tech ([Erik.Leppo@tetrattech.com](mailto:Erik.Leppo@tetrattech.com)).

\*If using the R code instead of the Shiny app, you won't need internet access if you have R installed on your computer and have downloaded all the necessary packages.

# How the app works

## Click buttons to

- Upload files
- Run the calculators
- Download results
- Unzip results



# Workflow

1. Import File # 1 (.csv file with required fields)
2. Go to 'Prepare Input File' tab and run the File Builder for the selected community (bugs or fish)
3. Download the results; unzip the file; resolve unmatched taxa if necessary
4. Import File # 2 (BCG\_TaxaTranslator\_Results\_wAttributes.csv output from the File Builder)
5. Run BCG calculation for the selected community (bugs or fish)
6. Download the results; unzip the file; interpret results

# Time saver tip

Some internet browsers, like Google Chrome and Mozilla Firefox, allow you to change settings so that you can select which folder to download files to (versus files going to the default Download folder).

- Mozilla Firefox: go to Settings > General > Files and Applications > Check box for “always ask where to save files.”
- [Google Chrome file saving tip](#) [PDF]

# Step 1: Set up a directory

This is the folder where you'll be saving your files



# Step 2: Prepare Input File

The File Builder joins the taxon name in the input file with the 'original taxalD' column in the [Taxa Translator table](#) and performs two functions:

- Translates the original taxon name to the proper Operational Taxonomic Unit (OTU) for the calculation (for example, collapses worms to Oligochaeta).
- Adds in new columns with taxa attributes and hierarchy (e.g., Order, Family) that are needed for the metric calculations.

[BioMonTools\\_SupportFiles/data/taxa\\_official/LowerBoise at main · leppott/BioMonTools\\_SupportFiles · GitHub](#)

# File Builder input file – required content

**Required fields for all calculations.** Column names are flexible (use drop-down menus to match).

| Column Name | Fish | Bugs | Data Type | Allowable entries                              | Description                                              |
|-------------|------|------|-----------|------------------------------------------------|----------------------------------------------------------|
| SampleID    | x    | x    | text      | flexible                                       | Unique sample identifier                                 |
| TaxaID      | x    | x    | text      | flexible                                       | Taxon name                                               |
| Count       | x    | x    | numeric   | non-negative real numbers; should not be blank | Number of individuals                                    |
| Index_Class | x    | x    | text      | see list below (must match exactly)            | Tells R which rules and metrics to apply, based on class |
| Length      | x    |      | numeric   | non-negative real numbers; blanks are OK       | Length, units = mm                                       |

**OPTIONAL** fields can be included as well and will be carried through into the output file.

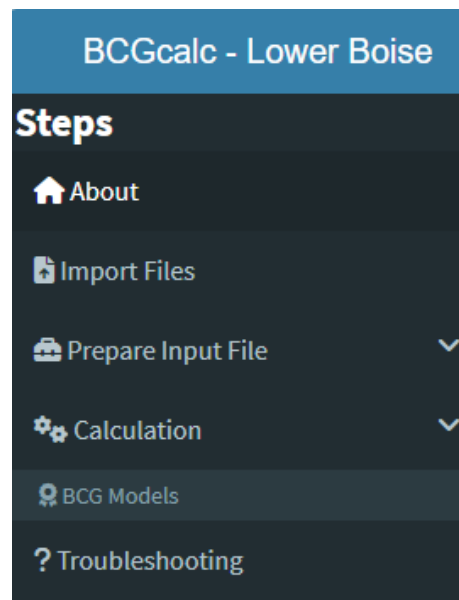
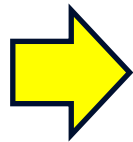
If you include DELT anomalies (fish) and density (bugs) the column names must match what is shown below.

| Column Name | Fish | Bugs | Data Type | Allowable entries                        | Description                                                                                            |
|-------------|------|------|-----------|------------------------------------------|--------------------------------------------------------------------------------------------------------|
| N_Anomalies | x    |      | numeric   | non-negative real numbers; blanks are OK | number of individuals with Deformity, Erosion, Lesion or Tumor (DELT) anomalies                        |
| Density_m2  |      | x    | numeric   | non-negative real numbers; blanks are OK | number of individuals per meter squared (count x subsample correction factor x area correction factor) |



# Step 3: Import File # 1

Must be a comma-separated file (CSV). It can include **data for multiple samples, sites and Index Classes** (versus being limited to one sample at a time).  
File uploads are limited to a maximum of 200 MB in size.



**Load File**  
Only comma-separated or tab-separated files.

Select file parameters

Separator  
☒ Comma  
☐ Tab

Choose file to upload  
Browse... BugBCG\_FileBuilder\_Test1\_20240725.csv  
Upload complete

The 'separator' allows the user to upload different file formats (e.g., csv, tsv, or txt).  
Files for all operations will be uploaded through this interface.  
File uploads are limited to a maximum of 200 MB in size.

A table is shown below after data is loaded.

Show 5 entries

Table. Imported data.

|   | SampleID | Taxonomic.Name          | Count | Index_Class | Index_Name |
|---|----------|-------------------------|-------|-------------|------------|
|   | All      | All                     | All   | All         | All        |
| 1 | Test1    | Acerpenna pygmaea       | 4     | bug4        | MN_BCG     |
| 2 | Test1    | Atherix variegata       | 1     | bug4        | MN_BCG     |
| 3 | Test1    | Brachycentrus numerosus | 4     | bug4        | MN_BCG     |
| 4 | Test1    | Caenis                  | 5     | bug4        | MN_BCG     |
| 5 | Test1    | Calopteryx aequabilis   | 3     | bug4        | MN_BCG     |

Showing 1 to 5 of 257 entries

Previous 1 2

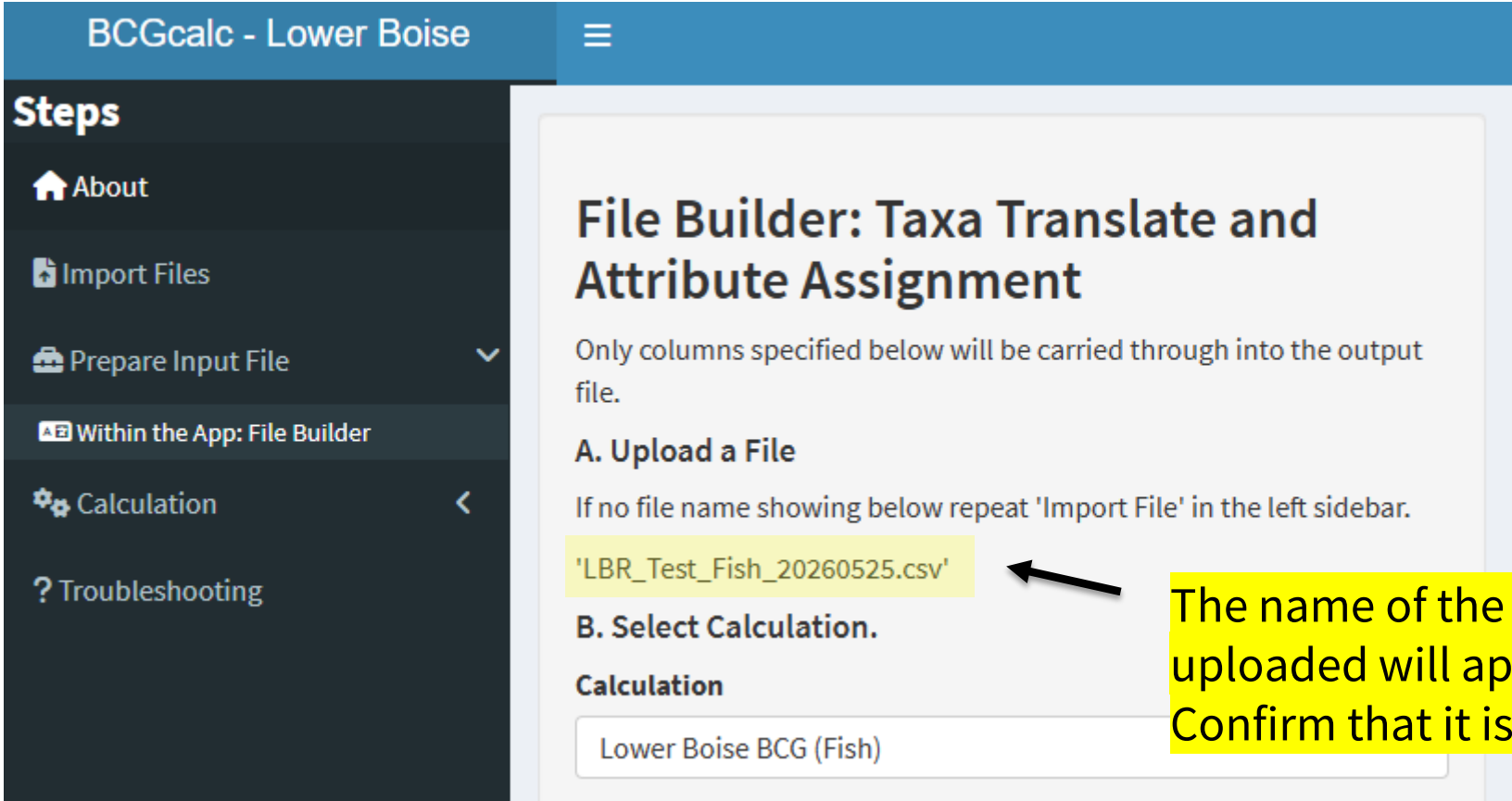
**1. Browse to the file**

**2. Status bar will tell you when the upload is complete**

**3. A file preview will appear. Check to make sure everything looks ok.**

# Step 4: Run File Builder

**IMPORTANT!** The BCG models are calibrated for specific collection and processing protocols. Make sure you are running the BCG calculator on appropriate data.



BCGcalc - Lower Boise

**Steps**

- About
- Import Files
- Prepare Input File
- Within the App: File Builder**
- Calculation
- Troubleshooting

### File Builder: Taxa Translate and Attribute Assignment

Only columns specified below will be carried through into the output file.

**A. Upload a File**

If no file name showing below repeat 'Import File' in the left sidebar.

'LBR\_Test\_Fish\_20260525.csv'

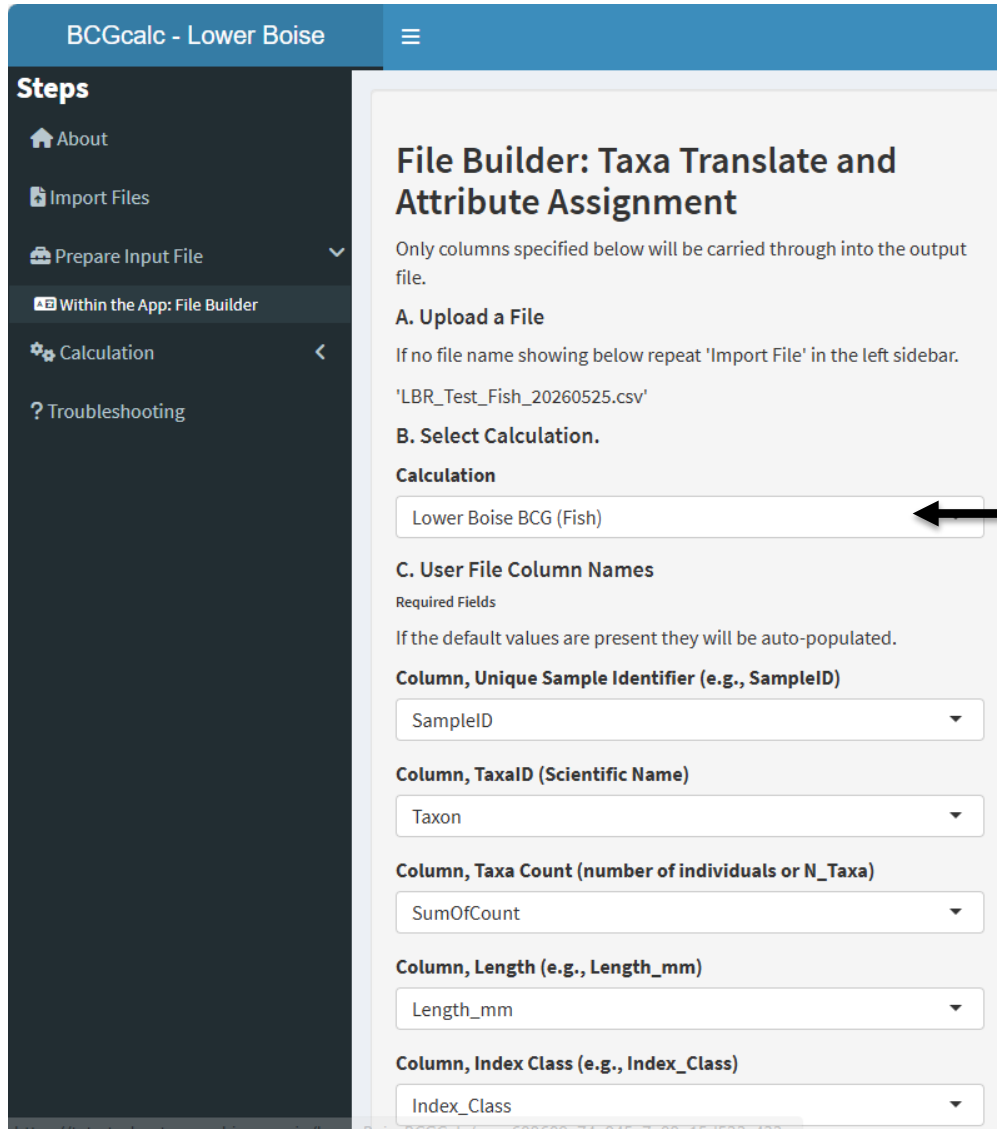
**B. Select Calculation.**

**Calculation**

Lower Boise BCG (Fish)

The name of the file that you uploaded will appear here. Confirm that it is the correct file.

# File Builder – enter required fields



The screenshot shows the 'File Builder: Taxa Translate and Attribute Assignment' interface. On the left is a sidebar with 'Steps' including About, Import Files, Prepare Input File, Within the App: File Builder (selected), Calculation, and Troubleshooting. The main panel has a title bar 'BCGcalc - Lower Boise' and a menu icon. The main content area is titled 'File Builder: Taxa Translate and Attribute Assignment' and includes instructions: 'Only columns specified below will be carried through into the output file.' It is divided into three sections: A. Upload a File (with a file name input field), B. Select Calculation (with a dropdown menu), and C. User File Column Names (with five dropdown menus for Required Fields). The dropdowns are currently set to 'Lower Boise BCG (Fish)', 'SampleID', 'Taxon', 'SumOfCount', and 'Length\_mm' respectively. A URL is visible at the bottom of the sidebar.

## Side note:

- **INDEX\_NAME** (*LBR\_Bugs\_BCG\_v1* or *LBR\_Fish\_BCG\_v1*) is also a required field but is autopopulated by R when you run the File Builder.

Select calculation

Enter the **required fields**

Drop-down boxes allow for flexibility with column names (match with the appropriate columns in the input file).

# File Builder – optional fields

**Columns to Keep in Output**

N\_ANOMALIES |

Index\_Class

SampleID

Taxon

SumOfCount

Length\_mm

All input and output files will be available in a single zip file.

 Download Results

**OPTIONAL** fields can be included as well and will be carried through into the File Builder output file.

If you include **DELT anomalies** (fish), the column name must be **N\_Anomalies**.

If you include **density (bugs)**, the column name must be **Density\_m2**.

# File Builder – Run Operation

Index\_Class ▼

Select Additional Required Fields (which vary by Calculation – see table on right).

Columns need to be named exactly as shown on the right. Optional fields (e.g., SiteID, collection date) can be selected as well. IMPORTANT! Do not include the same field twice and do not include Life Stage or other fields that might cause a taxon to occur in more than one row for a given sample (which could lead to double-counting of that taxon in the richness metrics).

**Columns to Keep in Output**

GP.RR

**D. Run Operation**

This button will merge the user file with the official taxa file

Run Operation

**E. Output**

All output files will be available in a single zip file.

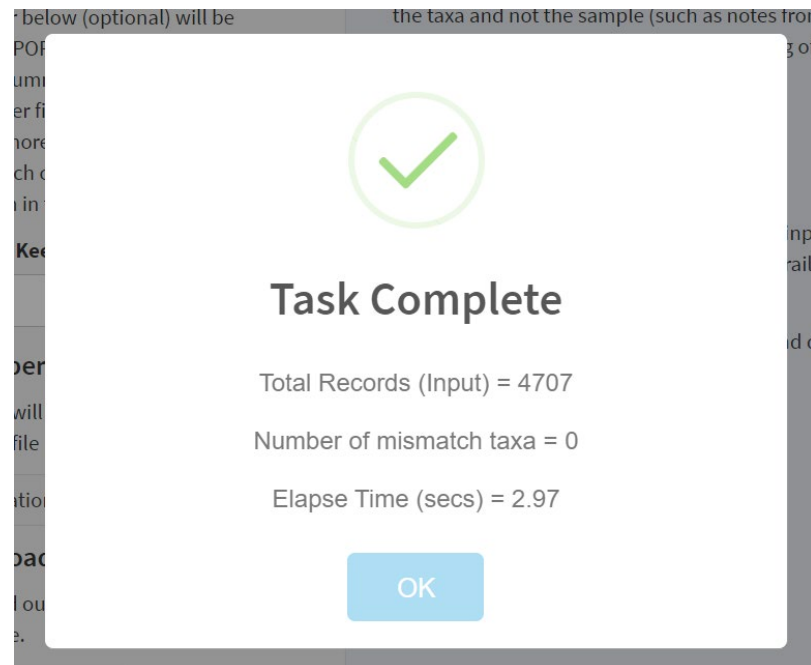
Download Results

Click on the Run Operation button

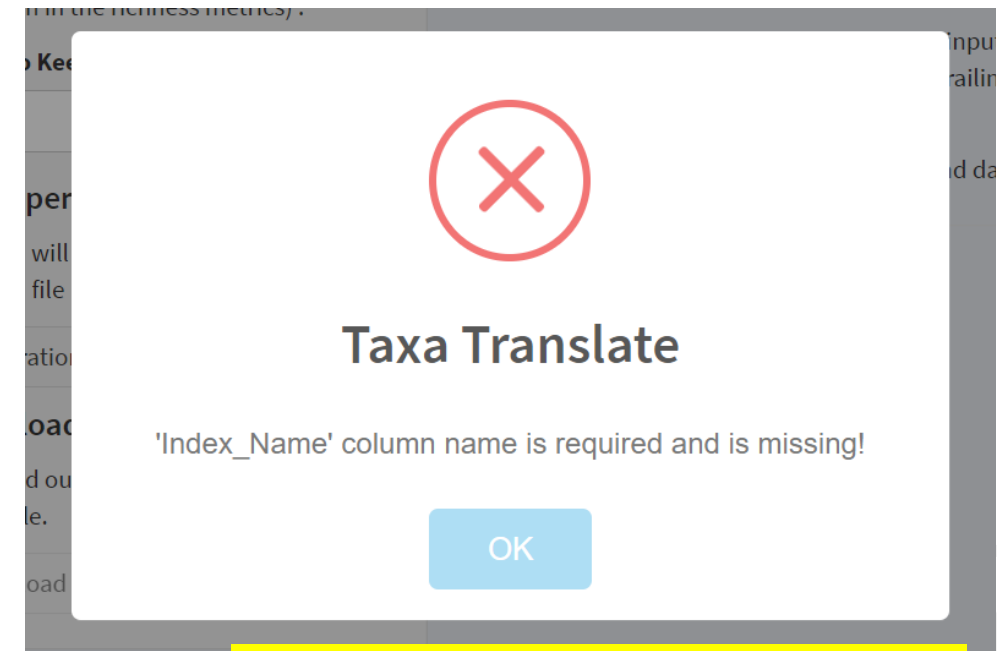
# File Builder – prompt box

When the File Builder has finished running, a **prompt box will appear**. It will notify you if your File Builder run was successful or failed.

Success!



Failure!

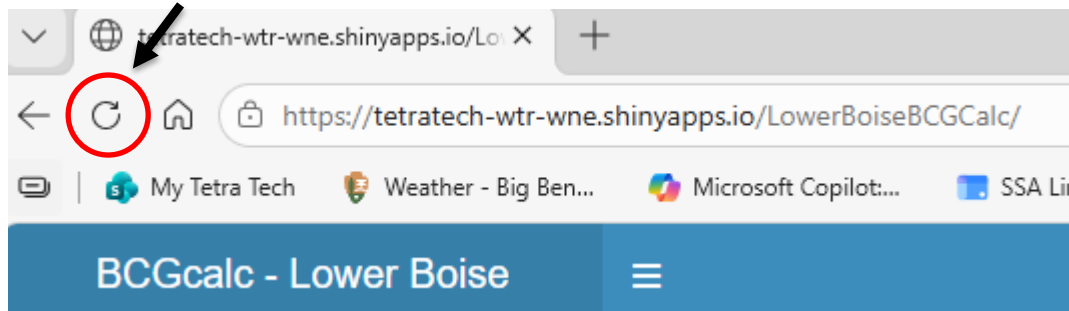


Will alert you to missing columns

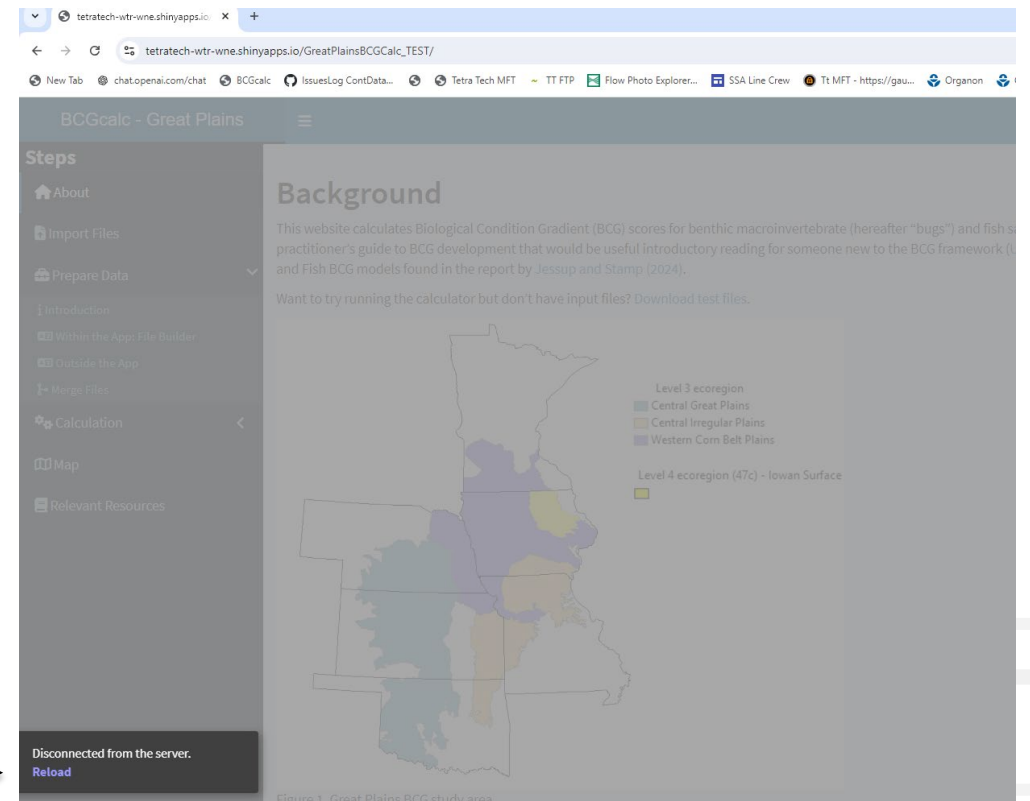
# Refresh the Shiny app

In some cases, the screen will gray out (meaning the Shiny app has crashed or timed out) and you'll have to refresh or reload the app, investigate potential problems with your input file, and try again.

Refresh button (upper left corner)



Reload if app crashes  
(gray screen)



# File Builder – Download Results

Index\_Class ▾

Select Additional Required Fields (which vary by Calculation – see table on right).

Columns need to be named exactly as shown on the right. Optional fields (e.g., SiteID, collection date) can be selected as well. IMPORTANT! Do not include the same field twice and do not include Life Stage or other fields that might cause a taxon to occur in more than one row for a given sample (which could lead to double-counting of that taxon in the richness metrics).

**Columns to Keep in Output**

GP.RR

**D. Run Operation**

This button will merge the user file with the official taxa file

Run Operation

**E. Download Output**

All input and output files will be available in a single zip file.

Download Results

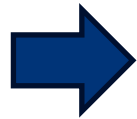


The 'Download Results' button will become clickable when the files are ready to download. Click the button and save the zipped file to the directory. The file will either automatically go to the Download folder or, if you use the Time Saver tip with Google Chrome or Mozilla Firefox (see slide 6), you can browse to the desired folder and save it there.



# Unzip File Builder results

Unzip  
the file



A 'results'  
folder will  
appear



Subfolders within the 'results' folder



\_user\_input\_Bugs



\_user\_input\_Fish



reference



results\_Bugs\_BCG



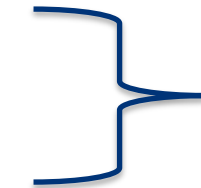
results\_Fish\_BCG



Original input files



Taxa translator, taxa attribute and fish age  
class files



File Builder & Calculation results

**Users must unzip/extract the files or else the files won't be seen as an option to upload for the next step. After files are unzipped, the original zipped files can be deleted if desired.**

# Unzipping files

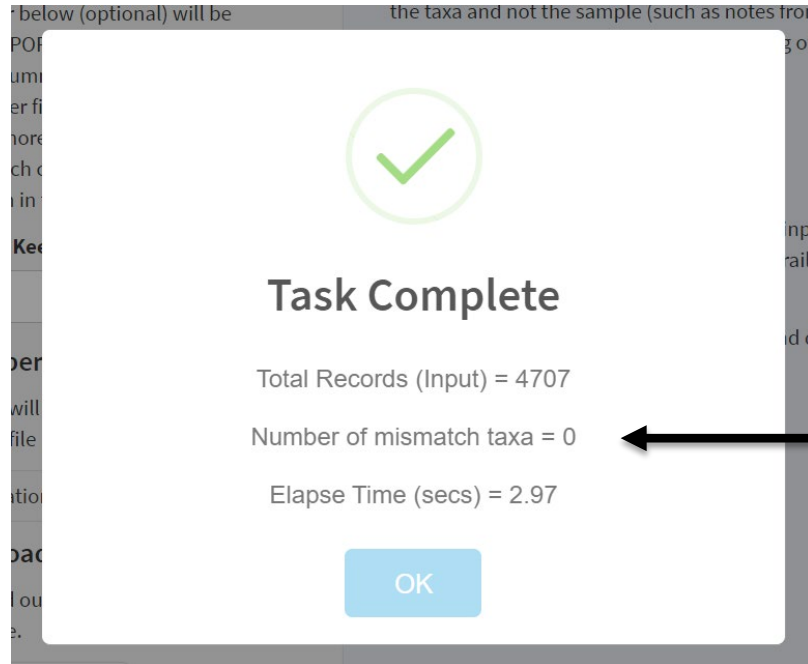
For whatever reason, 7zip (free) seems to work better than the Microsoft default zip extractor with our Shiny apps (but if you can't get 7zip, the Microsoft option should work ok).

7zip (<https://www.7-zip.org/>)

# File Builder outputs

| Output file                            | Description                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BCG_TaxaTranslator_modifications       | shows list of taxa in the input file; to see which taxa names were modified during the Taxa Translation process for reasons other than capitalization scheme, filter the Modified_woCAPS column = TRUE |
| BCG_TaxaTranslator_nonmatches          | taxa in your input file that don't match with the taxa list in the Taxa Translator table                                                                                                               |
| BCG_TaxaTranslator_Results_wAttributes | input file is now ready to go! The taxa names have been translated to the BCG OTU and attributes and hierarchy added in. Import this file into the Shiny app and run the BCG calculator                |
| BCG_TaxaTranslator_sources             | shows which calculation was selected, which Operational Taxonomic Unit (OTU) column was used, and which versions of the Taxa Translator and Attribute tables were used                                 |

# Non-matching taxa



## What if there are non-matching taxa?

Open the 'BCG\_TaxaTranslator\_nonmatch' file and view the list of non-matching taxa. If the non-matches are due to a misspelling, extra space or other small issue, fix the non-matching taxa in your input file, re-upload the file via the 'Import File' tab, and rerun the File Builder (and to prevent confusion, delete the first set of output files with the non-matching taxa). If the non-matching taxa are legitimate taxa that need to be added to the Taxa Translator table, notify [Jen.Stamp@tetratech.com](mailto:Jen.Stamp@tetratech.com) and Tetra Tech will update the Taxa Translator table as soon as resources permit.

Example



results\_Fish\_BCG



BCG\_TaxaTranslator\_modifications.csv

BCG\_TaxaTranslator\_nonmatches.csv

BCG\_TaxaTranslator\_Results\_wAttributes.csv

BCG\_TaxaTranslator\_sources.csv

# Step 5: Import File #2

Import the newly created 'BCG\_TaxaTranslator\_Results\_wAttributes' file

Example



results\_Fish\_BCG



BCG\_TaxaTranslator\_modifications.csv

BCG\_TaxaTranslator\_nonmatches.csv

BCG\_TaxaTranslator\_Results\_wAttributes.csv

BCG\_TaxaTranslator\_sources.csv

BCGcalc - Lower Boise



## Steps

About

Import Files

Prepare Input File

Calculation

Troubleshooting

### Load File

Only comma-separated or tab-separated files.

Select file parameters

#### Separator

☒ Comma

☐ Tab

#### Choose file to upload

Browse...

BCG\_TaxaTranslator\_Results\_wAttributes.csv

Upload complete

The 'separator' allows the user to upload different file formats (e.g., csv,

2. Status bar will tell you when the upload is complete

A table is shown below after data is loaded.

Show 5 entries

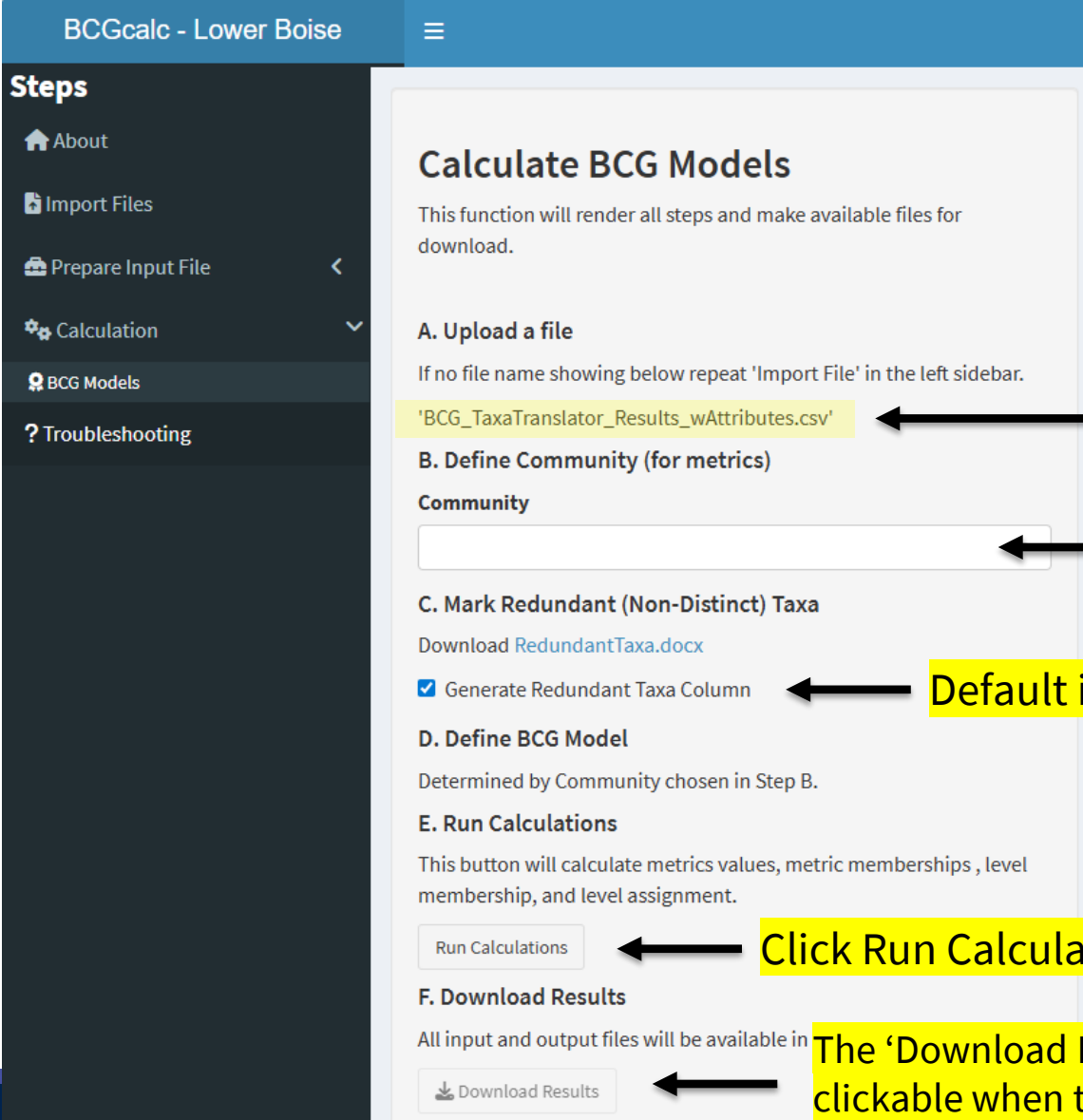
Search:

Table. Imported data.

|   | SampleID | TaxaID                   | N_Taxa | file_taxatrans                               | file_attributes                |
|---|----------|--------------------------|--------|----------------------------------------------|--------------------------------|
|   | All      | All                      | A      | All                                          | All                            |
| 1 | Test1    | Acari                    | 4      | LowerBoise_Bugs_Taxa_Translator_20260508.csv | LowerBoise_Bugs_Attributes_202 |
| 2 | Test1    | Acari                    | 8      | LowerBoise_Bugs_Taxa_Translator_20260508.csv | LowerBoise_Bugs_Attributes_202 |
| 3 | Test1    | Acari                    | 5      | LowerBoise_Bugs_Taxa_Translator_20260508.csv | LowerBoise_Bugs_Attributes_202 |
| 4 | Test1    | Acentrella insignificans | 2      | LowerBoise_Bugs_Taxa_Translator_20260508.csv | LowerBoise_Bugs_Attributes_202 |

3. A file preview will appear. Check to make sure everything looks ok.

# Step 6: Run desired calculation



The screenshot shows the BCGcalc - Lower Boise web application. A large yellow arrow points to the 'Calculation' step in the left sidebar. The main content area displays the 'Calculate BCG Models' section, which includes several steps and options. Annotations with arrows point to specific elements:

- The name of the file that you uploaded will appear here. Confirm that it is the correct file.** (Points to the file name: 'BCG\_TaxaTranslator\_Results\_wAttributes.csv')
- Select model** (Points to the 'Community' dropdown menu).
- Default is for R/Shiny app to mark redundant taxa** (Points to the checked checkbox 'Generate Redundant Taxa Column').
- Click Run Calculation button** (Points to the 'Run Calculations' button).
- The 'Download Results' button will become clickable when the files are ready to download** (Points to the 'Download Results' button).

The left sidebar contains the following steps: About, Import Files, Prepare Input File, Calculation (selected), BCG Models, and Troubleshooting. The main content area includes the following sections: A. Upload a file, B. Define Community (for metrics), C. Mark Redundant (Non-Distinct) Taxa, D. Define BCG Model, E. Run Calculations, and F. Download Results.

## BCG calculator workflow

- ✓ Mark non-distinct taxa (excluded from richness metrics)
- ✓ Calculate metric values each sample
- ✓ Calculate metric membership values for each sample
- ✓ Assign membership in each BCG level

# Unzip Calculation results

Unzip  
the file



Output files will  
be added to the  
existing folders



Subfolders within the  
'results' folder



results\_Bugs\_BCG



results\_Fish\_BCG



 \_BCG\_RESULTS.csv  
 \_BCG\_RESULTS.html  
 BCG\_1markexcl.csv  
 BCG\_2metval\_BCG.csv  
 BCG\_2metvall\_all.csv  
 BCG\_3metmemb.csv  
 BCG\_3metmemb\_xtab\_METRICSORT.csv  
 BCG\_3metrules.csv  
 BCG\_4levmemb.csv  
 BCG\_5levassign.csv  
 BCG\_6levflags.csv  
 BCG\_6metflags.csv  
 BCG\_TaxaTranslator\_modifications.csv  
 BCG\_TaxaTranslator\_nonmatches.csv  
 BCG\_TaxaTranslator\_Results\_wAttributes.csv  
 BCG\_TaxaTranslator\_sources.csv

Calculation  
output files

File  
Builder  
files

**After files are unzipped, the original zipped files can be  
deleted if desired.**

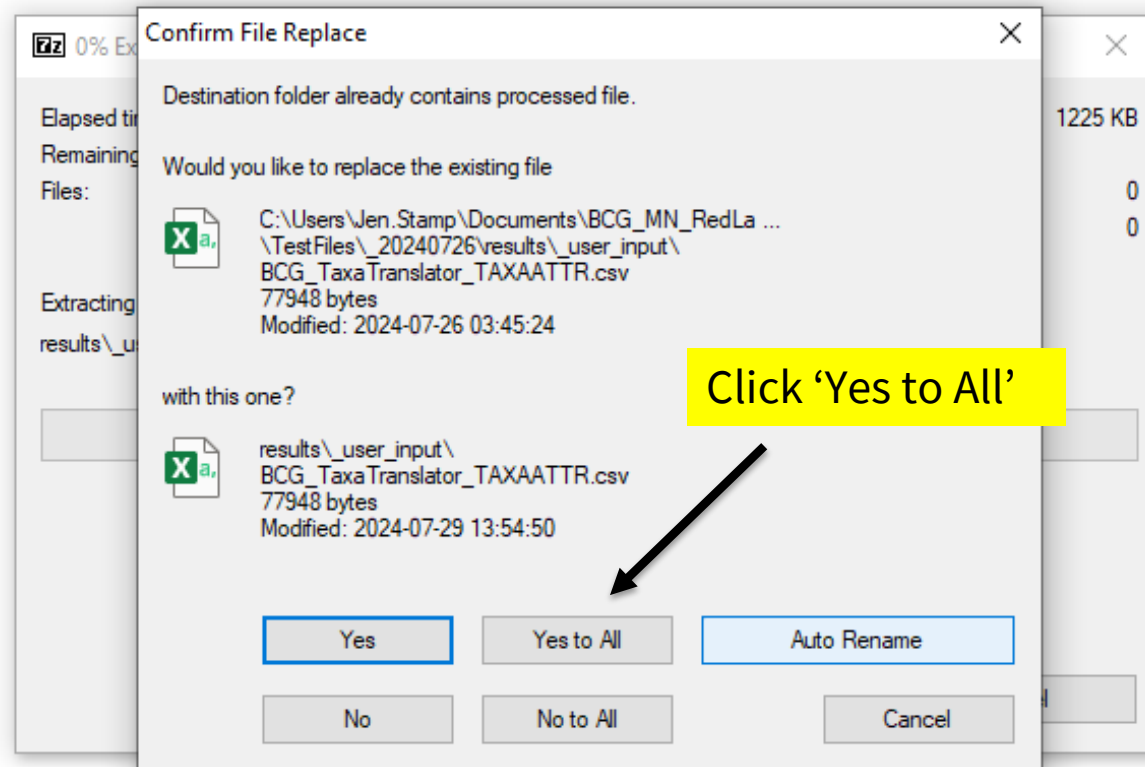
Fish outputs will have two additional outputs

- Fish\_AgeClass\_xtab
- Fish\_AgeClass\_pClasses



# Unzip Calculation results

You may receive a prompt like this when you unzip the Calculation results file.



## Questions? Comments?

We're always looking for feedback on ways we can improve the Shiny app.

Jen.Stamp@tetrattech.com

